Integration: Reversing the Chain Rule Part 1

Questions 1–21

Question 1. Find

a) $\int (x+1)^3 \, dx$
b) $\int (x-4)^6 \, dx$
c) $\int 6(2+x)^7 \, dx$
d) $\int (10(x-1)^4 - 4(x-3)^5) \, dx$

Question 2. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = (x-3)^{3/4}$
b) $\frac{dy}{dx} = (9+x)^{-3}$
c) $\frac{dy}{dx} = \sqrt{x+7}$
d) $\frac{dy}{dx} = \frac{4}{\sqrt{5+x}}$

Question 3. The curve y has gradient function $\frac{dy}{dx} = \frac{6}{(x-2)^5}$ and passes through the point $(3, \frac{5}{2})$. Show that y can be written in the form $y = a + \frac{b}{c(x-2)^4}$ where a, b and c are integers.

Question 4. The curve y has gradient function $\frac{dy}{dx} = \sqrt[3]{4+x}$ and passes through the point $(4, 3)$. Show that y can be written in the form $y = a(4+x)^b + c$, where a and b are rational numbers, and c is an integer.

Question 5. Evaluate the following definite integrals:

a) $\int_2^3 (x-2)^8 \, dx$

b) $\int_{-6}^{-3} 24(6+x)^5 \, dx$

c) $\int_{-1}^{1} \sqrt{x+3} \, dx$

d) $\int_{-3}^{4} \frac{44}{\sqrt[3]{4+x}} \, dx$

Question 6. Find the exact value of:

a) $\lim_{\delta x \to 0} \sum_{x=-5}^{26} (6+x)^{3/5} \delta x$

b) $\lim_{\delta x \to 0} \sum_{x=19}^{84} \frac{4}{\sqrt[4]{x-3}} \delta x$

Question 7. Find

a) $\int (2x+1)^4 \, dx$

b) $\int (2-3x)^5 \, dx$

c) $\int 4(5+6x)^9 \, dx$

d) $\int (3(2x-3)^3 - 4(4-5x)^4) \, dx$

Question 8. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = (5x+9)^{7/3}$

b) $\frac{dy}{dx} = (6-x)^{-5}$

c) $\frac{dy}{dx} = \sqrt{10+3x}$

d) $\frac{dy}{dx} = \frac{20}{\sqrt[3]{1-x}}$

Question 9. The curve y has gradient function $\frac{dy}{dx} = \frac{6}{(3x-2)^2}$ and passes through the point $(0, 2)$. Show that y can be written in the form $y = \frac{ax+b}{cx+d}$, where a, b, c and d are integers.

Question 10. The curve y has gradient function $\frac{dy}{dx} = \frac{14}{(4x+1)^{9/2}}$ and passes through the point $(0, 2)$. Show that y can be written in the form $y = a + \frac{b}{(4x+1)^c}$, where a and b are integers and c is a rational number.

Question 11. Evaluate the following definite integrals:

a) $\int_2^3 (2x-5)^5\,dx$

b) $\int_{-1/4}^{1/8} \sqrt{3+8x}\,dx$

c) $\int_1^2 \frac{15}{(5-6x)^2}\,dx$

d) $\int_{5/16}^{3/2} \frac{8}{\sqrt[3]{3+16x}}\,dx$

Question 12. Find the exact value of:

a) $\lim_{\delta x\to 0} \sum_{x=-1/3}^{10} \frac{2}{5}(2+3x)^{3/5}\delta x$

b) $\lim_{\delta x\to 0} \sum_{x=1/9}^{1/3} \frac{243}{(4-9x)^4}\delta x$

Question 13. Find

a) $\int 2x(x^2+3)^5\,dx$

b) $\int 12x^2(4x^3-5)^3\,dx$

c) $\int x(2x^2-5)^4\,dx$

d) $\int 2x^3\sqrt{x^4-3}\,dx$

Question 14. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = 9x^2(3x^3+2)^{-2}$

b) $\frac{dy}{dx} = 10x^3(2-5x^4)^{5/2}$

c) $\frac{dy}{dx} = x\sqrt[3]{3+4x^2}$

d) $\frac{dy}{dx} = \frac{100x^8}{(2x^9+3)^6}$

Question 15. The curve y has gradient function $\frac{dy}{dx} = 5x(5x^2 - 1)^7$ and passes through the point $(0, 1)$. Show that y can be written in the form $y = a(5x^2 - 1)^b + c$, where a and c are rational numbers, and b is an integer.

Question 16. The curve y has gradient function $\frac{dy}{dx} = \frac{9x^2}{\sqrt{27x^3 - 1}}$ and passes through the point $(\frac{1}{3}, 0)$. Show that y can be written in the form $y = a(27x^3 - 1)^b$, where a and b are rational numbers.

Question 17. Evaluate the following definite integrals:

a) $\int_0^2 x(10x^2+1)^4\,dx$

b) $\int_{1/\sqrt{2}}^{3/2} x(4x^2-1)^{2/3}\,dx$

c) $\int_2^3 \frac{6x^2}{(x^3+2)^2}\,dx$

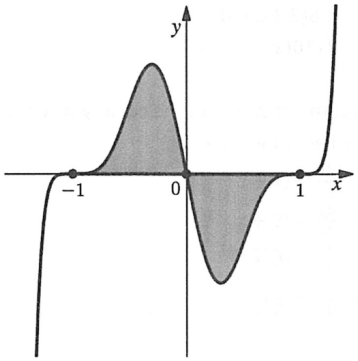
d) $\int_0^{1/2} \frac{800x^3}{(1-x^4)^3}\,dx$

Question 18. Find the exact value of:

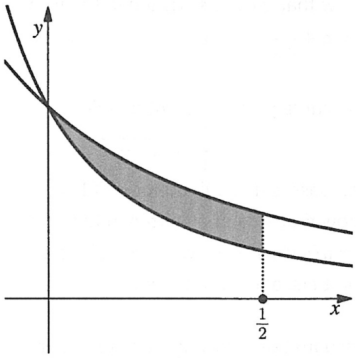
a) $\lim_{\delta x\to 0}\sum_{x=0}^{1} 98x(1+14x^2)^{5/2}\delta x$

b) $\lim_{\delta x\to 0}\sum_{x=0}^{9} \frac{\sqrt{\sqrt{x}+1}}{\sqrt{x}}\delta x$

Question 19. The diagram below shows the curve with equation $y = \frac{1}{2}x(x^2 - 1)^5$. Find the exact value of the shaded area.



Question 20. The diagram below shows the curves $y = \frac{2}{(x+1)^2}$ and $y = \frac{2}{(2x+1)^2}$. Find the exact shaded area between the two curves.



Question 21. The diagram below shows the curve with equation $y = \frac{12x-9x^2}{(2x^2-x^3+1)^3}$. Find the exact value of the shaded area.

